



BITS Pilani
Pilani Campus

Module 1: Properties and Handling of Particulate Solids (Chapter 28)

Lecture 6

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LECTURE: EQUIPMENT FOR SIZE REDUCTION

Classification of Size Reduction Equipment



Method by which force applied

Impact

- Impact at one surface
- Impact between particles

Compression

- Crushing
- Grinding

Rubbing/Attrition between Surfaces

Shear/Cutting of the solid

Nonmechanical Introduction of Energy

- Thermal Shock
- Explosive Shattering
- Electrohydraulic crushing
- Cryogenic Crushing
- Ultrasonic Crushing

Size-Reduction Equipment

Based on Size of Feed and Product

Crushers

- Do the heavy work of breaking large pieces of solid material into small lumps

Primary Crusher

- Operates on run-of-mine material
- Accepting anything that comes from the run-of-mine face and breaking it into 150- to 250-mm lumps.

Secondary Crusher

- A secondary crusher reduces lumps from primary crusher to particles perhaps 6 mm in size.

Grinders

- Reduce crushed feed to powder

Intermediate Grinder

- From an **intermediate grinder** might pass a 40-mesh screen;

Fine Grinder

- most of the product from a **fine grinder** would pass a 200-mesh screen with a 74- μ m opening.

Ultrafine Grinders

- accepts feed particles no larger than 6 mm; the product size is typically 1 to 50 μ m.

Cutting Machines

- give particles of definite size and shape
- 2 to 10 mm in length.

Principal types of size-reduction machines



A. Crushers (coarse and fine)

1. Jaw crushers
2. Gyratory crushers
3. Crushing rolls

B. Grinders (intermediate and fine)

1. Hammer mills; impactors
2. Rolling-compression mills
 - a. Bowl mills
 - b. Roller mills
3. Attrition mills
4. Tumbling mills
 - a. Rod mills
 - b. Ball mills; pebble mills
 - c. Tube mills; compartment mills

C. Ultrafine grinders

1. Hammer mills with internal classification
2. Fluid-energy mills
3. Agitated mills
4. Colloid mills

D. Cutting machines

1. Knife cutters; dicers; slitters

Selection criteria of size-reduction equipment



- It should produce the materials of desired shape and size or the size distribution desired
- It should accept the maximum input size expected.
- It should have a large capacity
- It should not choke or plug
- It should pass unbreakable materials without causing damage to itself
- It should operate economically with minimum supervision and maintenance
- The power input per unit weight of product should be small
- It should resist abrasive wear
- It should be dependable and have prolonged service life
- The replacement parts should be readily available at cheaper price
- The initial fixed cost and the operating cost should be minimum
- It should be easy and safe to operate
- It should have easy access to internal parts for maintenance

- Types of crushers:

- Jaw Crushers
- Gyratory crushers
- Smooth-roll crushers
- Toothed-roll crushers

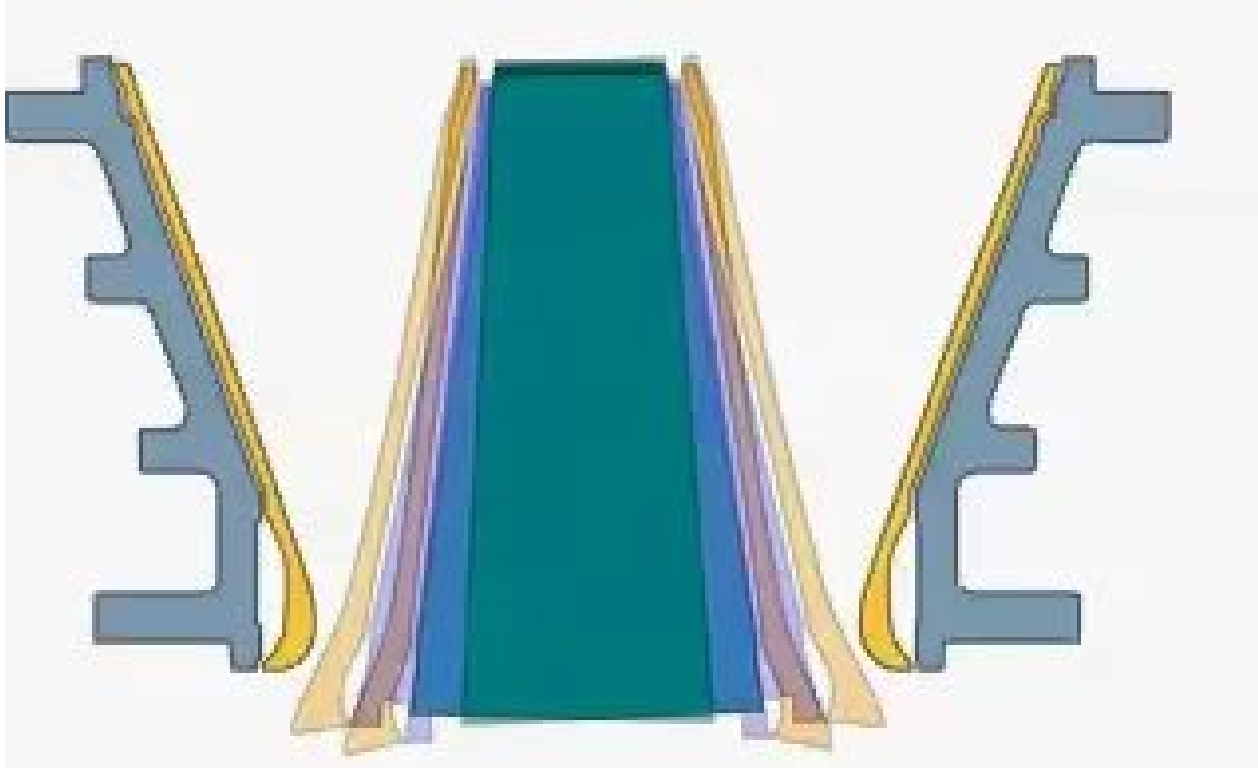
- Principle

- Compression
- Break large lumps of very hard materials

- Applications

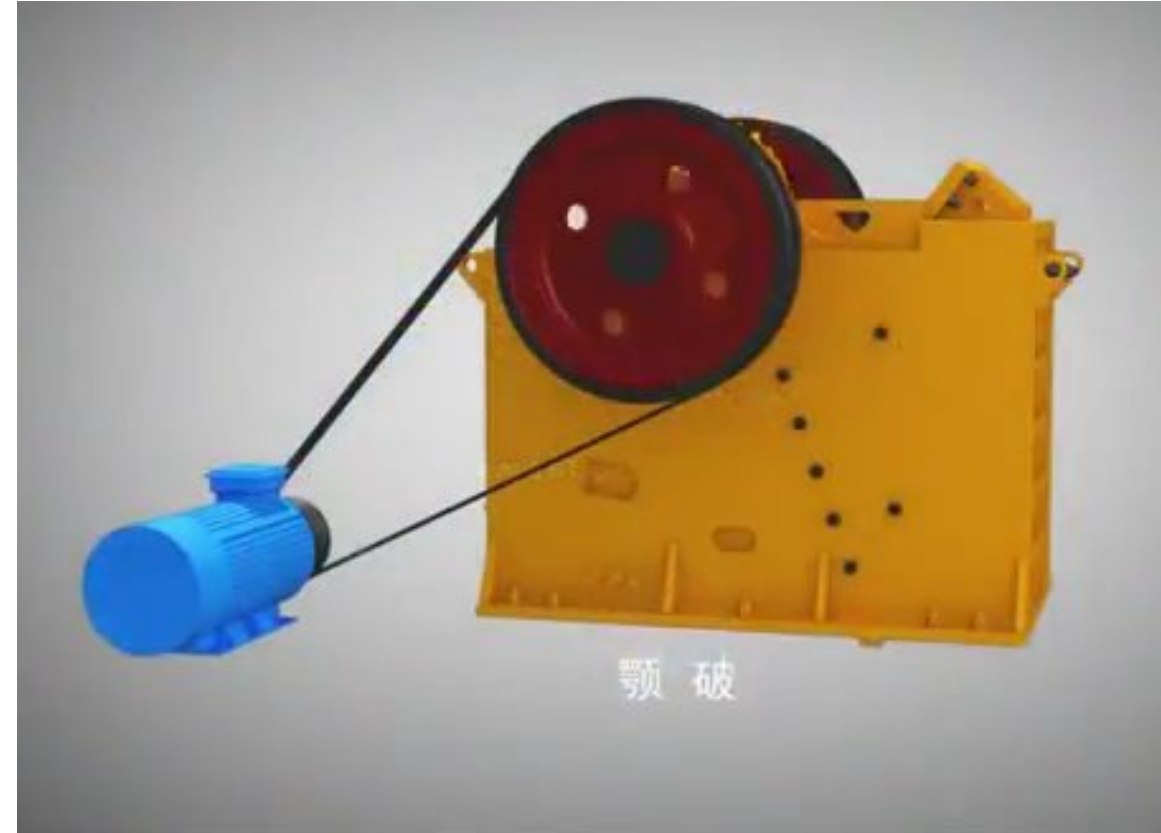
- Mining
- Cement manufacture
- Large-scale operations

➤Crusher Working Principle



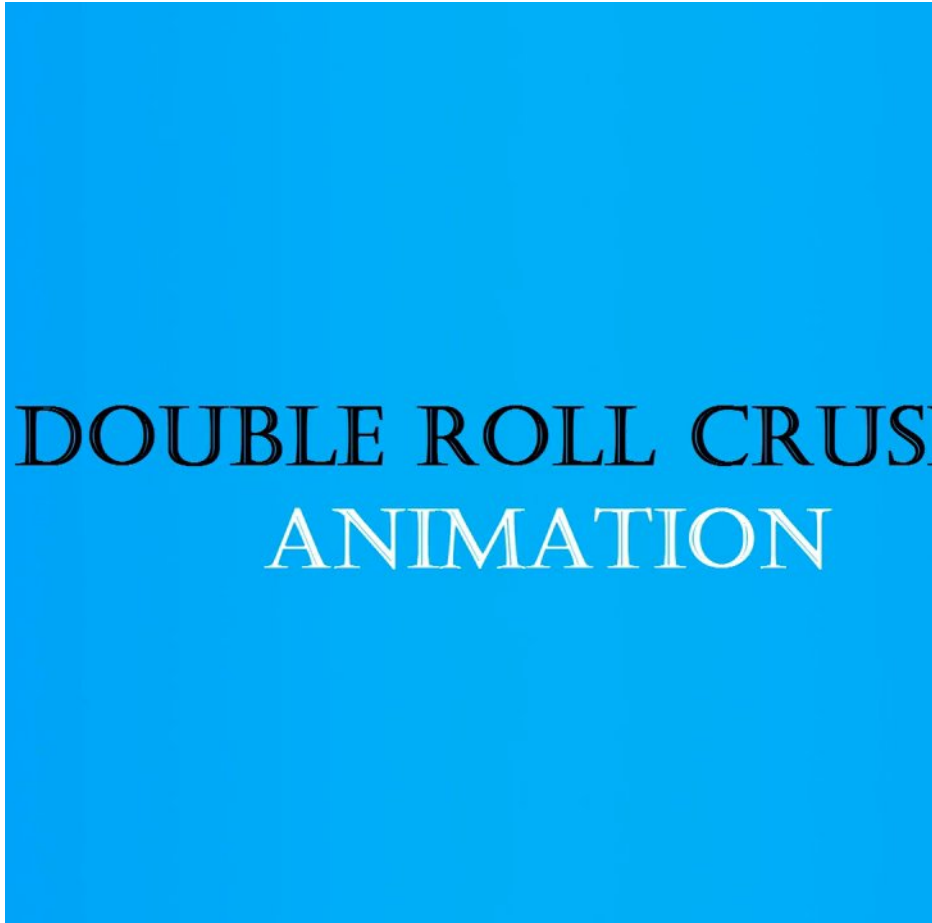
<https://youtu.be/VHojD6-tdCs>

➤Jaw Crusher



<https://www.youtube.com/watch?v=ECbBtcPkI8Y>

➤ Smooth Roll Crusher



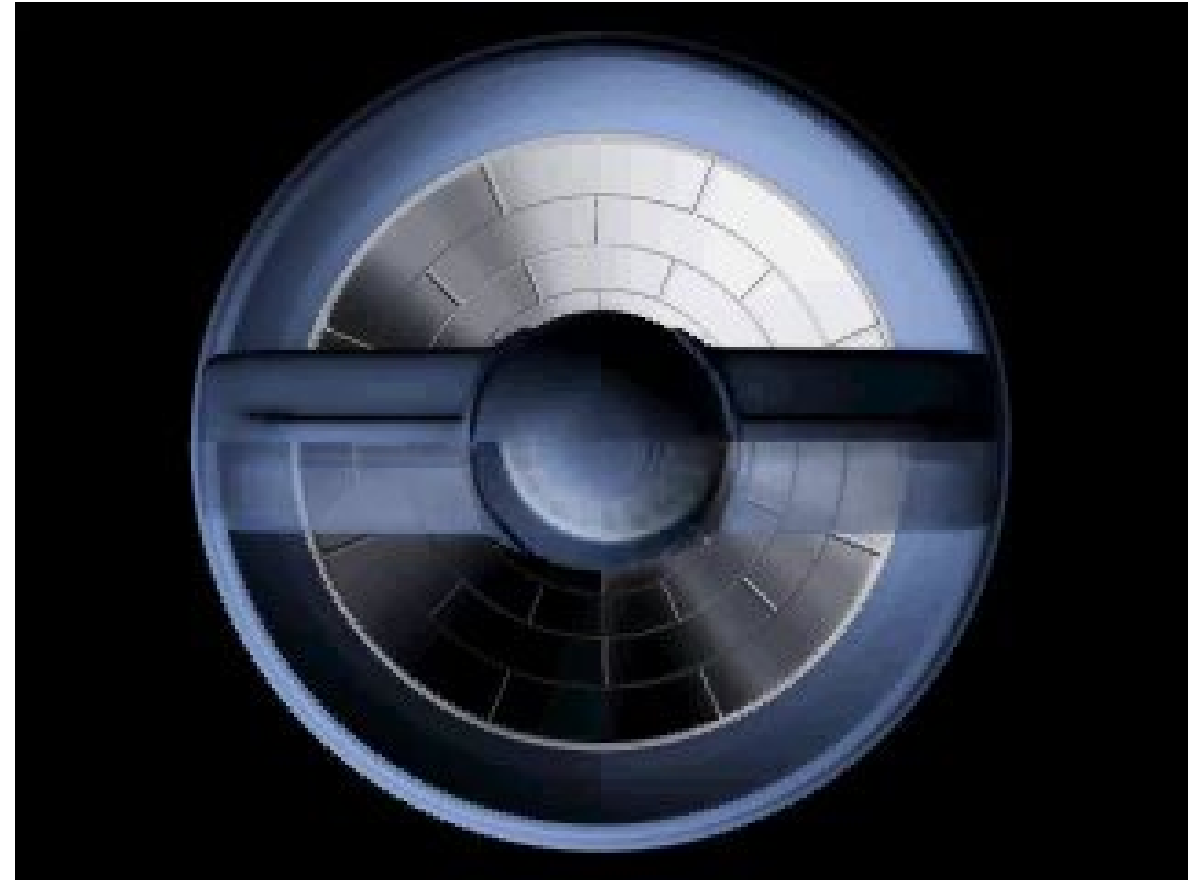
DOUBLE ROLL CRUSHER
ANIMATION

- These are secondary crushers producing a product 1 to 12 mm in size
- They are limited by the size of the particle that can be nipped by the rolls
- Feed ranges in size from 12 to 75 mm

<https://www.youtube.com/watch?v=6aMfcGfY5ko>

➤ Gyratory Crusher

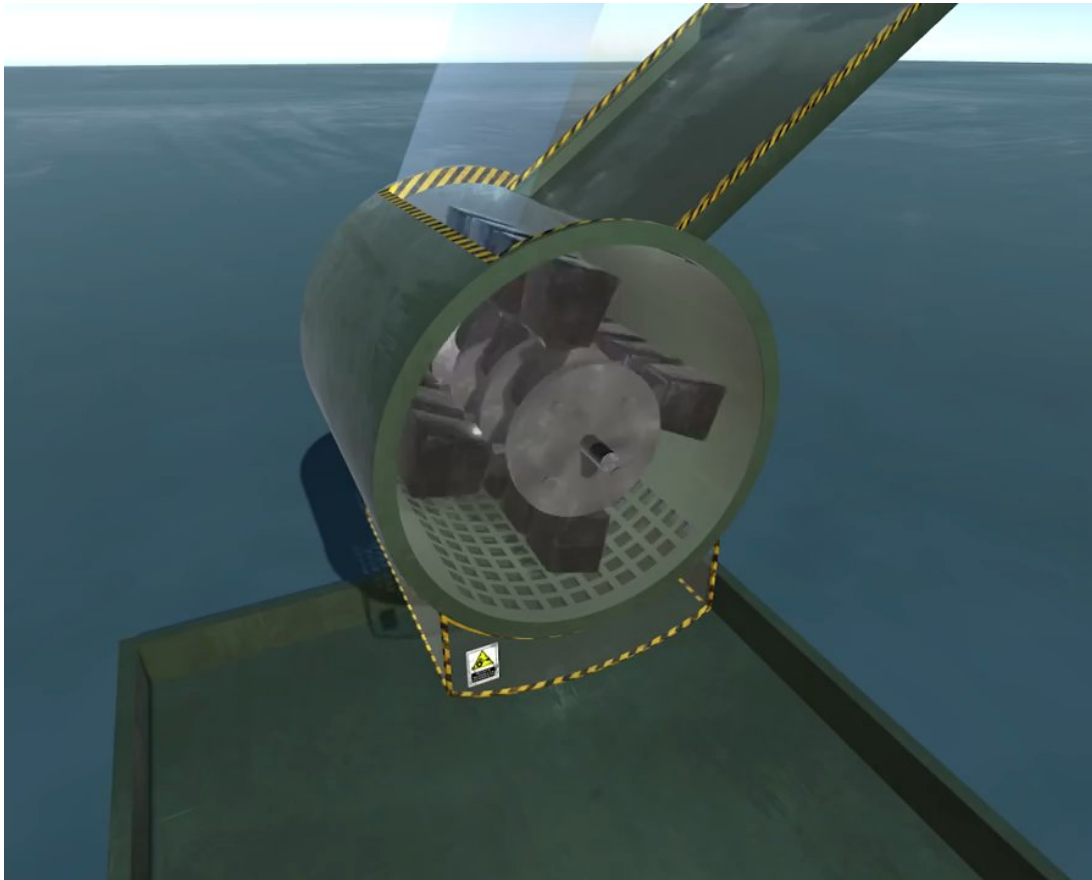
- A conical crushing head gyrates inside a funnel-shaped casing, open at the top
- An eccentric drives the shaft carrying the crushing head.
- Solids caught between the head and the casing are broken and rebroken until they pass out the bottom.



<https://youtu.be/IYeOW7UCZAg>

- Types:
 - Hammer mills and impactors
 - Rolling compression machines
 - Attrition mills
 - Tumbling mills (ball mill)
- Intermediate size reduction duty
- Product from crusher fed to grinder

➤ Hammer mills and impactors



- These mills contain high-speed rotor turning inside a cylindrical casing, usually the shaft is horizontal.
- Feed dropped into the top of the casing is broken and falls out through a bottom opening.
- The particles are broken by sets of swing hammers pinned to a rotor disk.
- Intermediate hammer mills yield a product 25 mm to 20-mesh in particle size.
- Hammer mills grind almost anything– tough fibrous solids like bark or leather, steel turnings, soft wet pastes, sticky clay, hard rock.
- For fine grinded they are limited to softer materials.

<https://youtu.be/fSTvbE77ucE>

➤ Roller mills

- In Roller mills solids are caught and crushed between vertical cylindrical rollers and a stationary anvil ring or bull ring.
- The rotors are driven at moderate speeds in a circular path.
- Plows lift the solid lumps from the floor of the mill and direct them between the ring and rolls.
- A stream of air sweeps the product to a classifier separator
- Oversized particles are returned to the mill for further reduction.
- Applications:
 - Reduction of limestone,
 - cement clinker, and
 - coal.

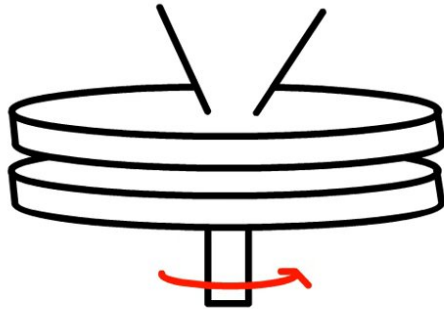
VERTICAL ROLLER MILL



<https://youtu.be/Am7TSGOOqZg>

➤ Attrition Mill

Grinding Equipments
Grinders (Burr mill)



<https://youtu.be/FmbnDdeBZ4o>

➤ Tumbler mills (ball mills)



<https://youtu.be/L6sgGXXYdEU>

- **Tumbler mills (ball mills)**
 - Most of the reduction is done by impact

- **Critical speed** of a ball mill in revolution per second:

$$n_c = \frac{1}{2\pi} \sqrt{\frac{g}{R-r}}$$

- g is the acceleration due to gravity, m/s^2
 - R is the radius of the mill, m
 - r is the radius of the grinding elements, m
- The operating speed must be less than critical speed
- Usually run at 65 to 80% of the critical speed, lower for wet grinding in viscous suspensions

- Many commercial powders must contain particles averaging 1 to 20 μm in size, with substantially all particles passing a standard 325-mesh screen that has openings 44 μm wide.
- Mills that reduce solids to such fine particles are called Ultra fine grinders.
- Ultrafine grinding of dry powder is done by grinders, such as high-speed hammer mills, provided with internal or external classification, and by fluid-energy or jet mills.
- Ultrafine wet grinding is done in agitated mills.



Fluidised Bed Opposed Jet Mill
www.jetmillmfg.com

- In these mills the particles are suspended in a high-velocity gas stream.
- In some designs the gas flows in a circular or elliptical path; in others there are jets that oppose one another or vigorously agitate a fluidized bed.
- Some reduction occurs when the particles strike or rub against the walls of the confining chamber, but most of the reduction is believed to be caused by interparticle attrition.
- Internal classification keeps the larger particles in the mill until they are reduced to the desired size.

- In some size-reduction problems the feed stocks are too tenacious or too resilient to be broken by compression, impact, or attrition.
- In other problems the feed must be reduced to particles of fixed dimensions.
- These requirements are met by devices that cut, chop, or tear the feed into a product with the desired characteristics.



<https://youtu.be/-IEUhwOCTLU>

- A crusher, grinder, or cutter cannot be expected to perform satisfactorily unless
 1. the feed is of suitable size and enters at a uniform rate;
 2. the product is removed as soon as possible after the particles are of the desired size;
 3. unbreakable material is kept out of the machine; and
 4. in the reduction of low-melting or heat-sensitive products, the heat generated in the mill is removed.

Open-Circuit and Closed-Circuit Operation



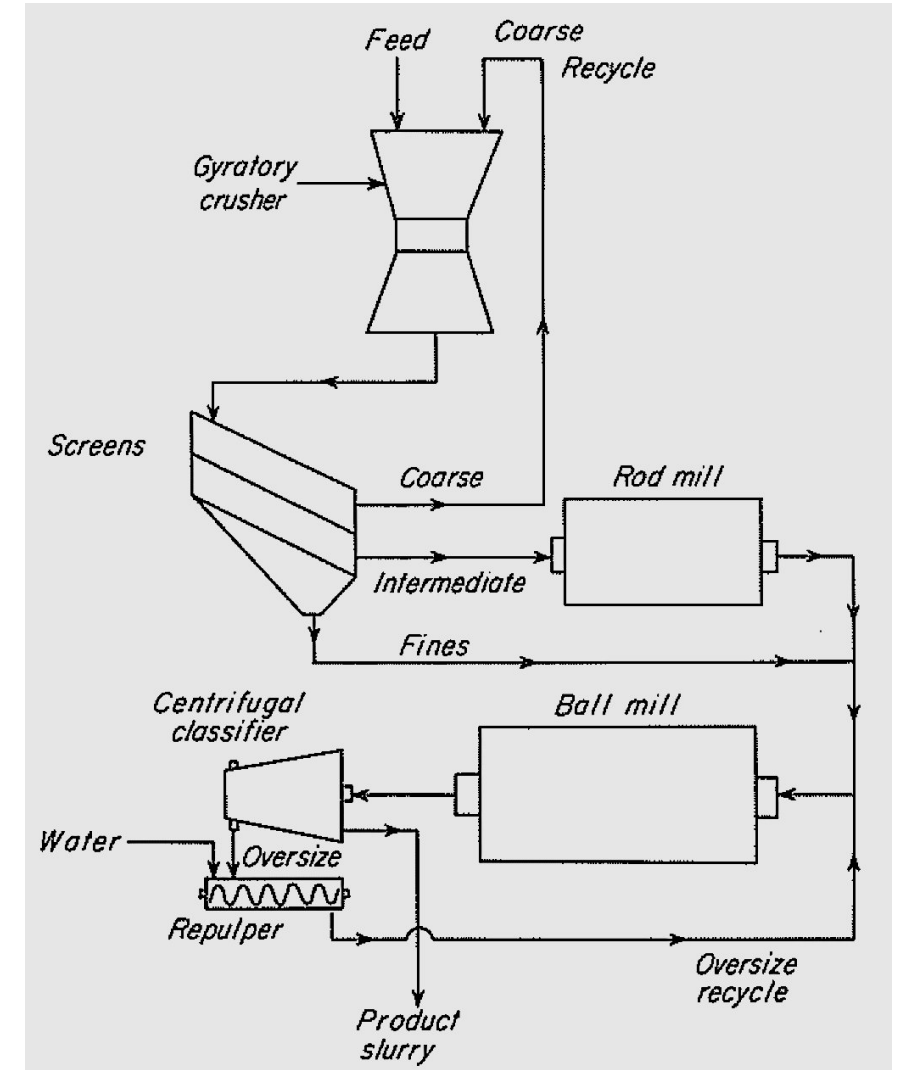
Open Circuit Operation:

- In many mills the feed is broken into particles of satisfactory size by passing it once through the mill.
- No attempt is made to return oversize particles to the machine for further reduction
- This may require excessive amounts of power, for much energy is wasted in regrinding particles that are already fine enough.

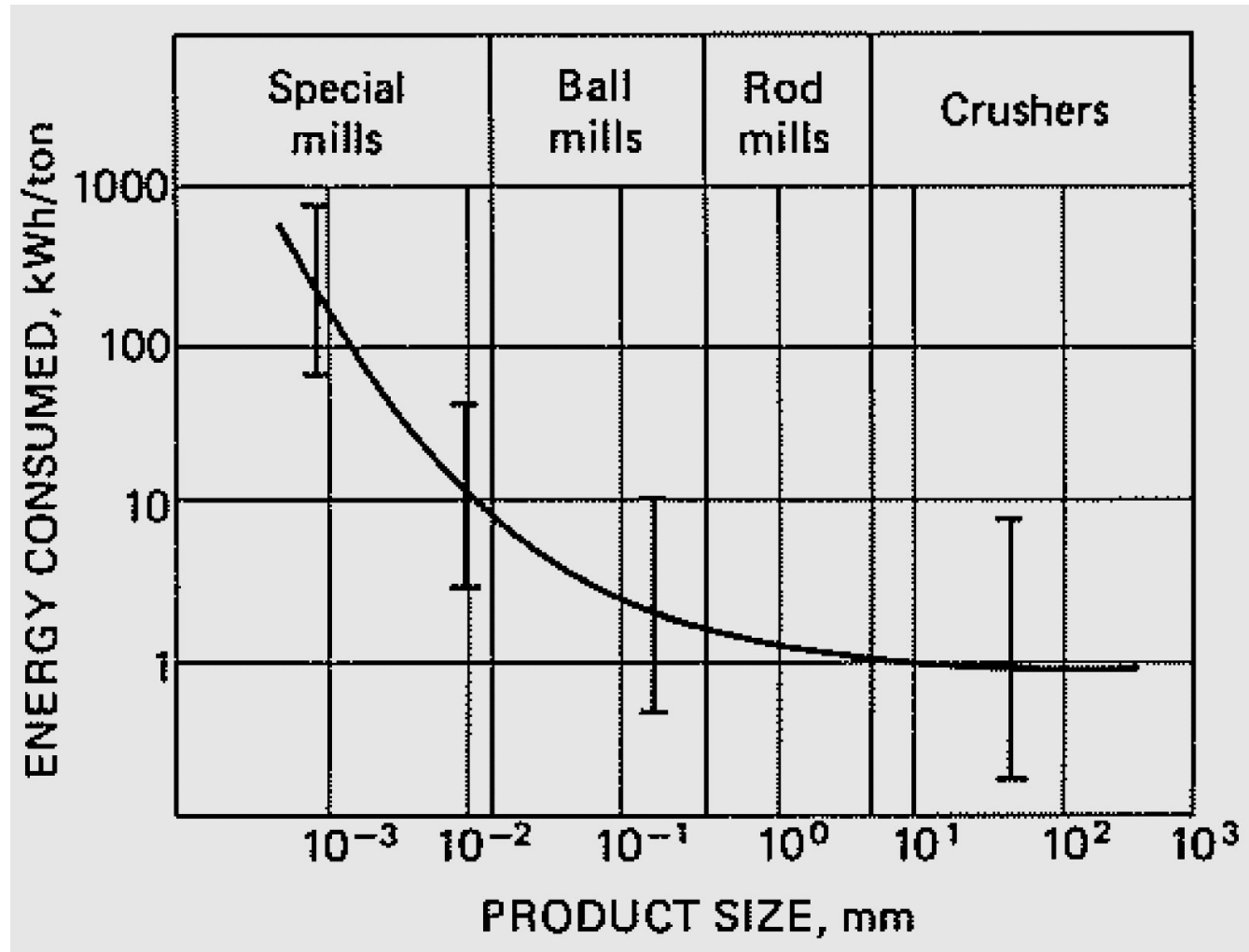
Closed-circuit operation

- the term applied to the action of a mill and separator connected so that the oversize particles are returned to the mill.

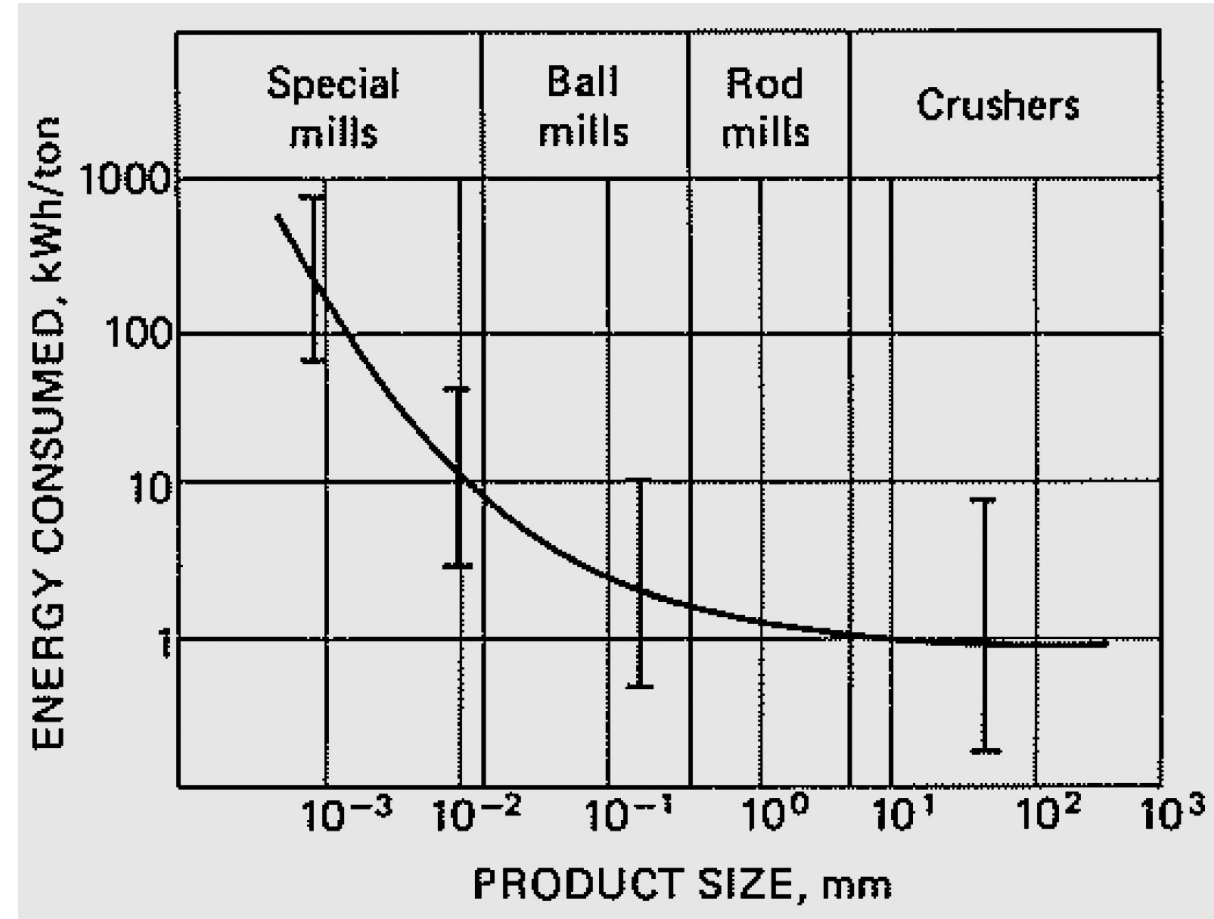
Flow sheet for Closed-circuit operation



Energy Consumption



- Size reduction is probably the most inefficient of all unit operations
- Over 99 percent of the energy goes to operating the equipment, producing undesirable heat and noise, leaving less than 1 percent for creating new surface.
- As processes have been developed that require finer and finer particles as feed to a kiln or reactor, the total energy requirement has increased, for reduction to very fine sizes is much more costly in energy than simple crushing to relatively coarse products.



- Only a very small fraction of the energy supplied to the solid is used in creating new surface, the bulk of the energy is converted to heat, which may raise the temperature of the solid by many degrees.
- The solid may melt, decompose, or explode unless this heat is removed.
- Cooling water or refrigerated brine is often circulated through coils or jackets in the mill.
- Sometimes the air blown through the mill is refrigerated, or solid carbon dioxide (dry ice) is admitted with the feed.
- Still more drastic temperature reduction is achieved with liquid nitrogen, to give grinding temperatures below -75°C .
- The purpose of such low temperatures is to alter the breaking characteristics of the solid, usually by making it more friable.

References



- Smith, J., Harriott, P., McCabe, W. (2005). Unit Operations of Chemical Engineering. Colombia: McGraw-Hill Education.
- Swain, A. K., Patra, H., & Roy, G. K. (2011). Mechanical operations. Tata McGraw Hill.